BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 093 · Lesson 093 · Von Neumann and Harvard Architectures

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does an address bus carry?

Q2. What does a data bus transfer?

Q3. Define contention.

Q4. A D flip-flop has D=1 and Q=0 immediately before a rising edge. What is Q just after?

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I Do · Read new knowledge

Key vocabulary

Von Neumann — Shared instruction/data memory organisation.

Harvard — Separate instruction/data stores or pathways.

bottleneck — Limiting shared resource.

modified Harvard — Modern mixed memory hierarchy.

New learning

Observe — Von Neumann stores programs and their data in a common memory/address space, so instruction fetches and data transfers may contend for shared resources.

Explain — Harvard architecture separates instruction and data pathways, potentially allowing simultaneous instruction fetch and data access.

Evaluate — Modern systems often use unified main memory but separate instruction/data caches; the classification is a model of design trade-offs, not a universal speed ranking.

Worked example

Explain one potential bottleneck of simple Von Neumann design.

1. Explain the mechanism — Instruction and data transfers share memory/bus resources, so competing accesses can wait. — Describe the shared resource and its effect, not claim all Von Neumann machines are slow.

We Do · Practise with guidance

A controller uses distinct code ROM and data RAM accessed via separate pathways. Which model?

Explain the mechanism

Harvard-style separation; fetch instruction and access data paths may operate in parallel.

It differs from a single shared path even when control logic is the same CPU.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State difference between Von Neumann and Harvard memory organisation. [4]

Question 2

Explain Von Neumann bottleneck with instruction fetch and data read. [6]

Question 3

Evaluate 'Harvard architecture always makes programs faster than Von Neumann' with contemporary caches and workload. [6]

Exit Ticket · Answer before checking

Contemporary processors can combine a unified main memory with separate instruction and data caches.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Memory location.

Q2. Content value/instruction bits.

Q3. Two operations compete for same limited resource.

Q4. Q becomes1, sampled at that active edge.

You Do pages · Independent questions

Q1. Von Neumann shared instruction/data storage/path [2], Harvard separate instruction/data memory/pathways [2].

Q2. Shared route [2], competing transfers [2], stalls or throughput effect [2].

Q3. Separate paths may reduce contention [2], real systems use modified Harvard caches [2], speed still depends on cache, clock and workload [2].

Exit Ticket · Model answer

True. This is a common modified-Harvard arrangement.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: